Processed Plates vs Planetary Peril

A cross-border nutritional analysis revealing the animal-plant divide as the primary driver of environmental footprints

Erdos Data Science Final Project | 2026

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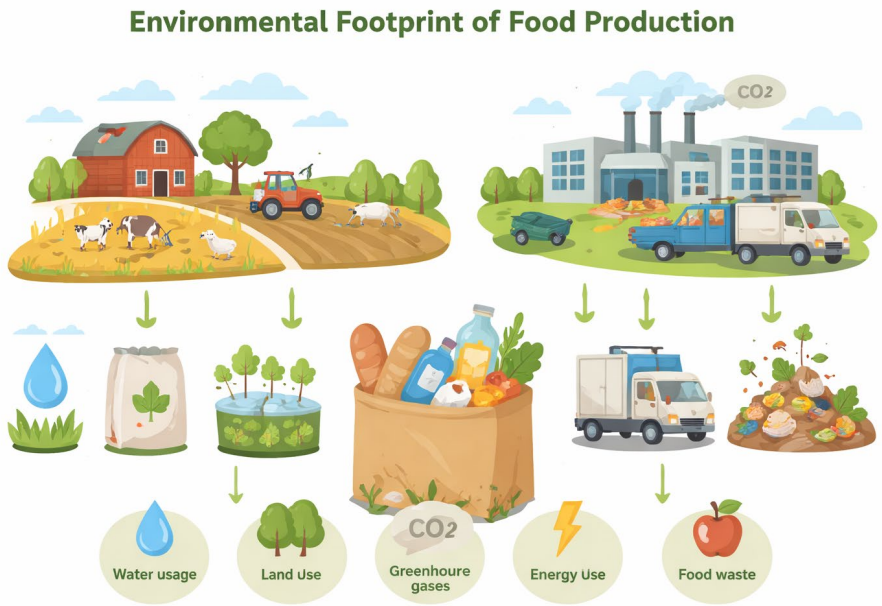
Motivation

- Ultra-processed foods linked to health issues (obesity, Type 2 diabetes and cardiovascular disease) [Lane et al., 2024, BMJ]
- Folk wisdom: Eat less processed food
- If ultra-processing is bad for human health, is it also bad for the environment?



- Nutrition labels are found universally – key to international portability
- Train a model using American data, and apply to French food data

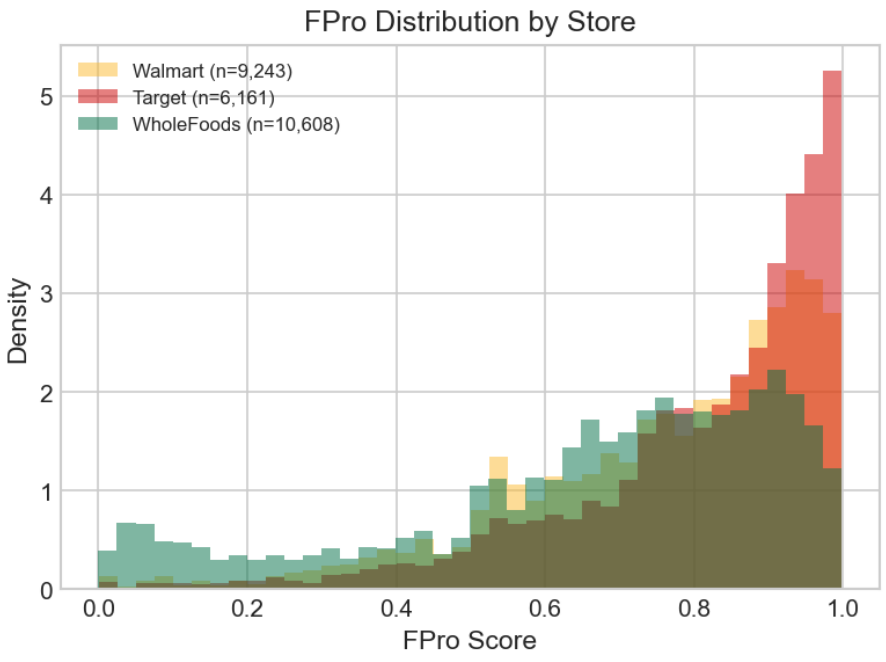
Data & Pipelines

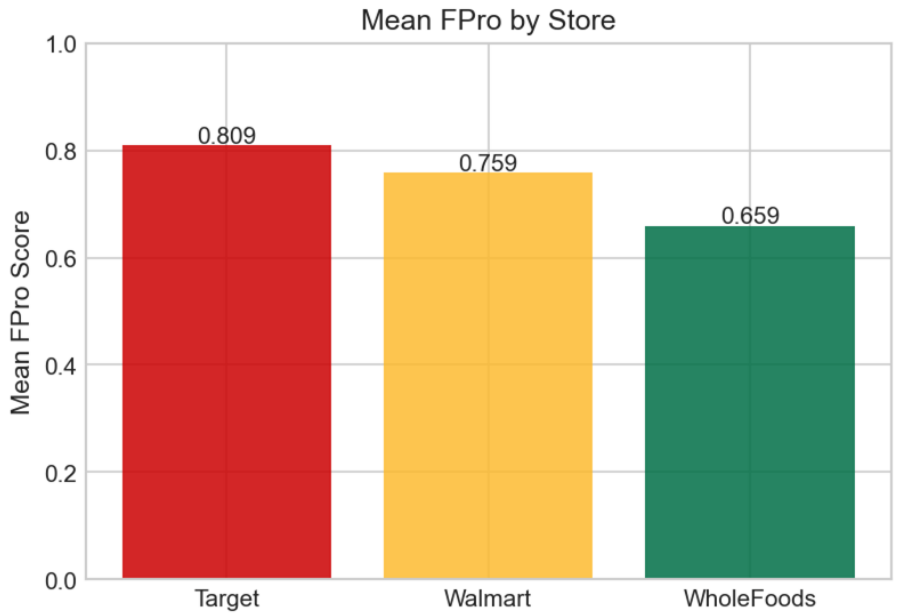
Datasets

- GroceryDB (US): 12,032 products with FPro + nutrition
- FPro scores between 0 and 1
- CIQUAL 2025 (France): 3,484 reference foods, with nutritional info
- Agribalyse 3.1 (France): 1,085 product-level LCA scores

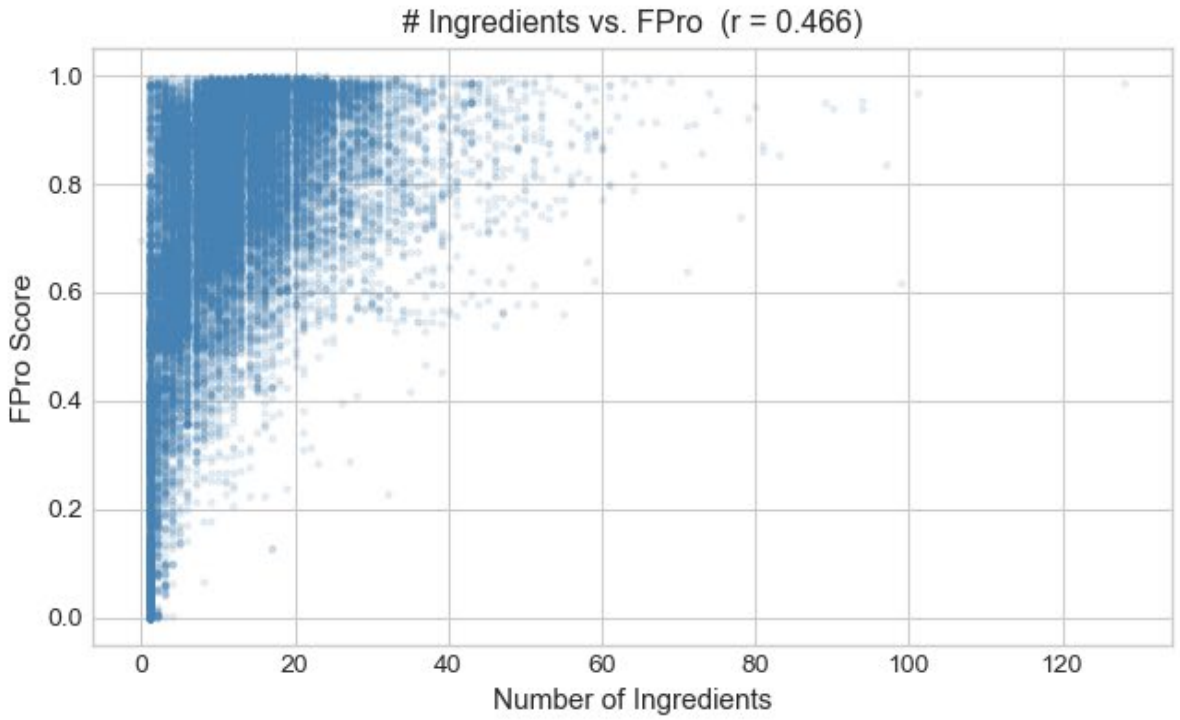
Pipeline

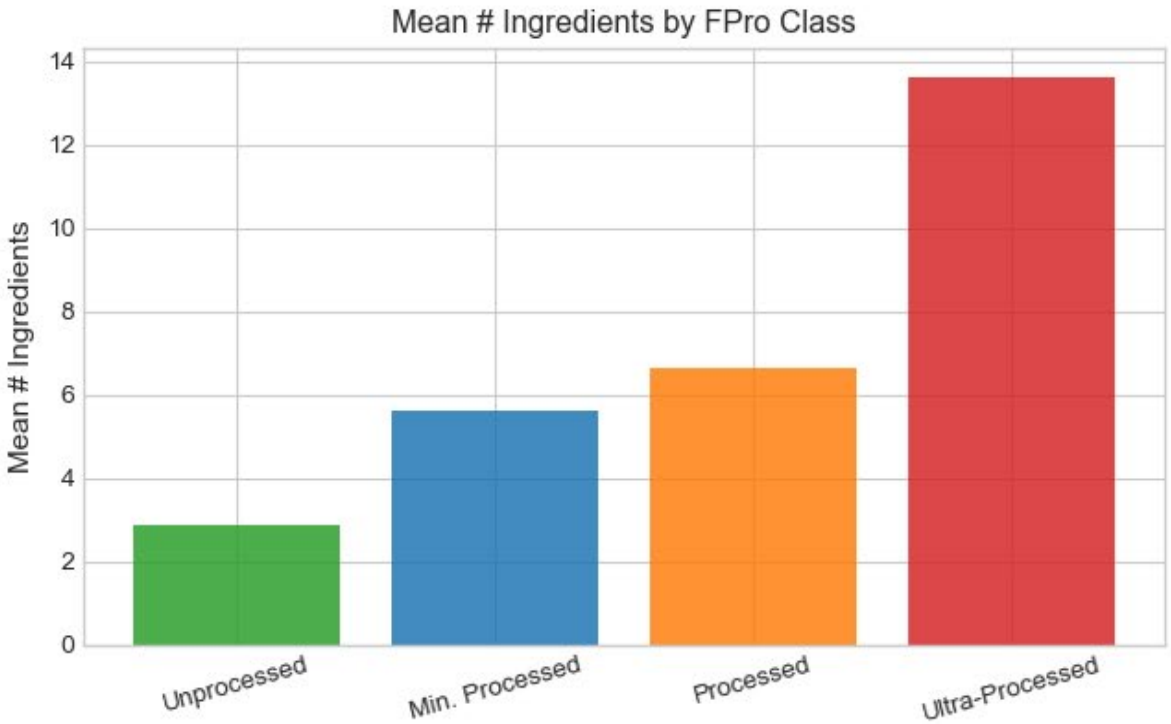
- Investigate using FPro scores and Environmental Footprint (EF)
- 12 nutrients → log-transform → Train models → FPro
- Apply to CIQUAL → Merge Agribalyse → Correlations





Number of ingredients vs FPro





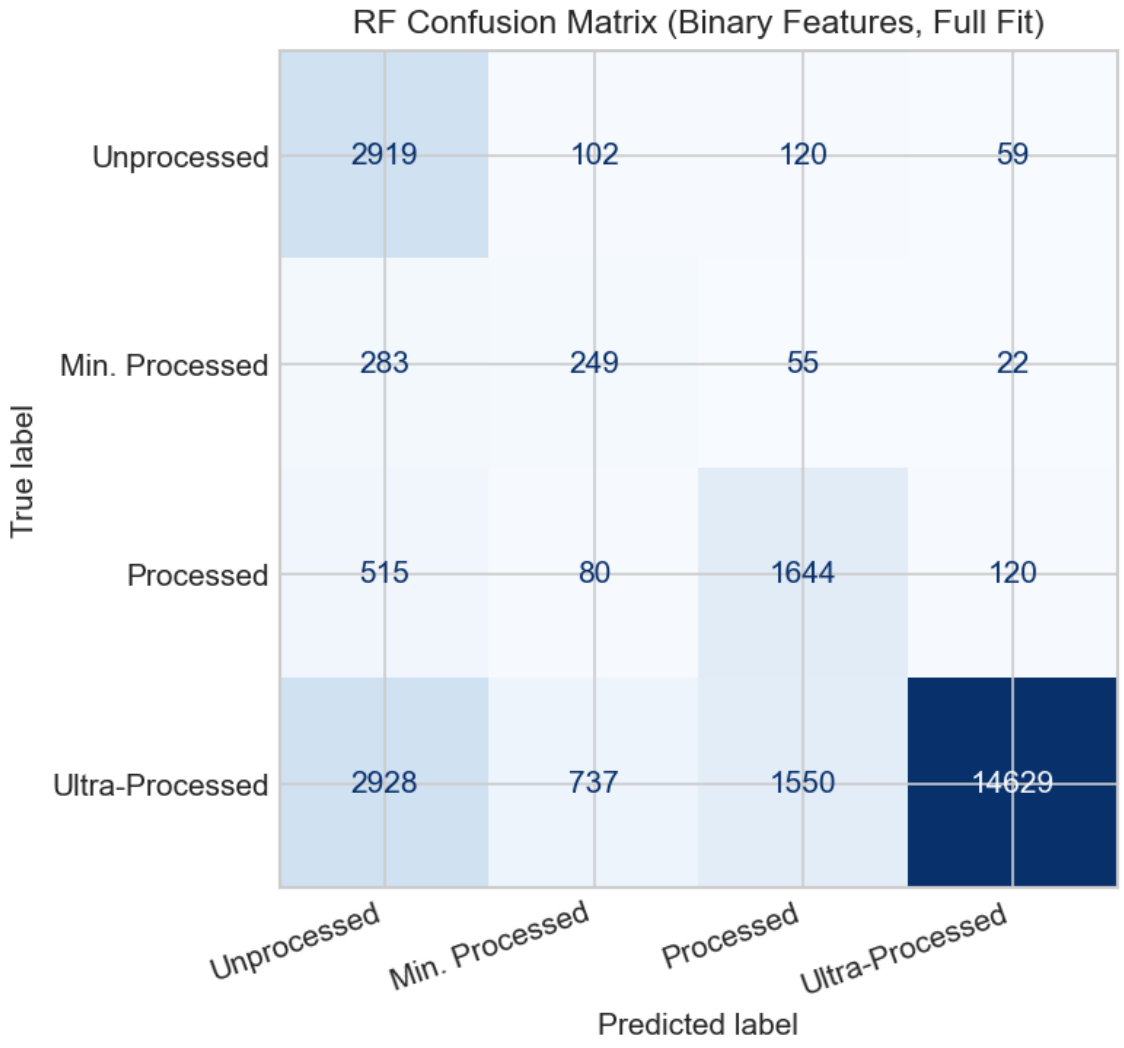
Pearson coefficient, r = 0.47

Moderate positive correlation: More ingredients = More processed

- Too simple. Need to see what the ingredients are!

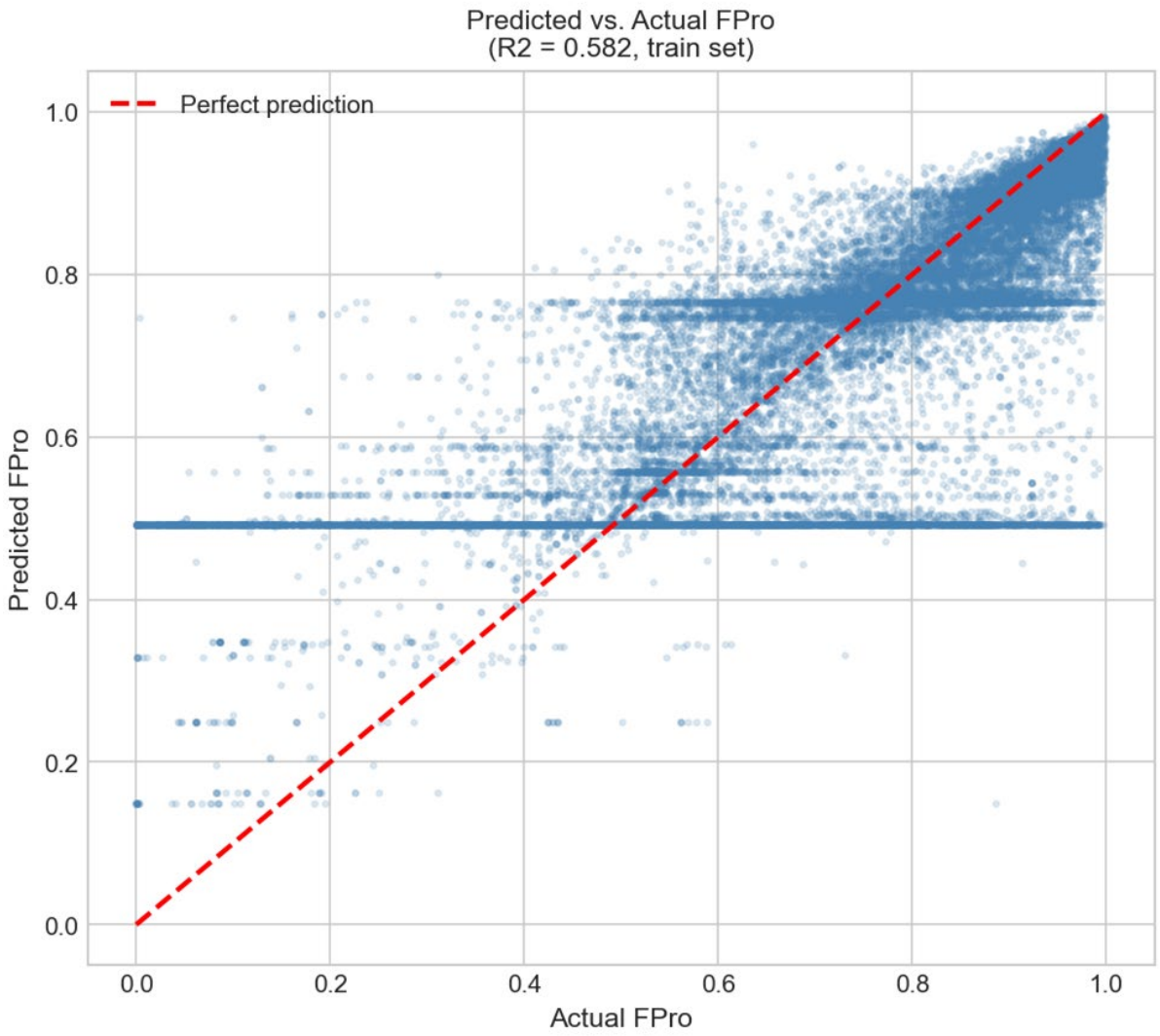
Ingredient (Binary)-FPro Classifier

- Can the presence/absence of the most common ingredients determine processing level (0-3)?
- Used a Random Forest classifier with Stratified K-fold CV
- Ultra-processed misclassified as unprocessed
- Example: Short ingredient lists but highly processed contents (high fructose corn syrup, soy lecithin)



Ingredient (Rank)-FPro Score

- Rank each ingredient by its position 1/rank (1st ingredient = 1.0, 2nd = 0.5, etc.), encoding both *presence* and *importance*
- Used Random Forest with K-fold CV
- Sugar and salt dominate — their position in the ingredient list is the strongest predictor of processing
- Rank is slightly better than binary (acc to Macro F1).
- Horizontal line: Too few ingredients match TOP_K feature list. Low signal → default to predict mean (0.5).



Hyperparameter tuning and Model comparison

Table 1: Regression Model Comparison

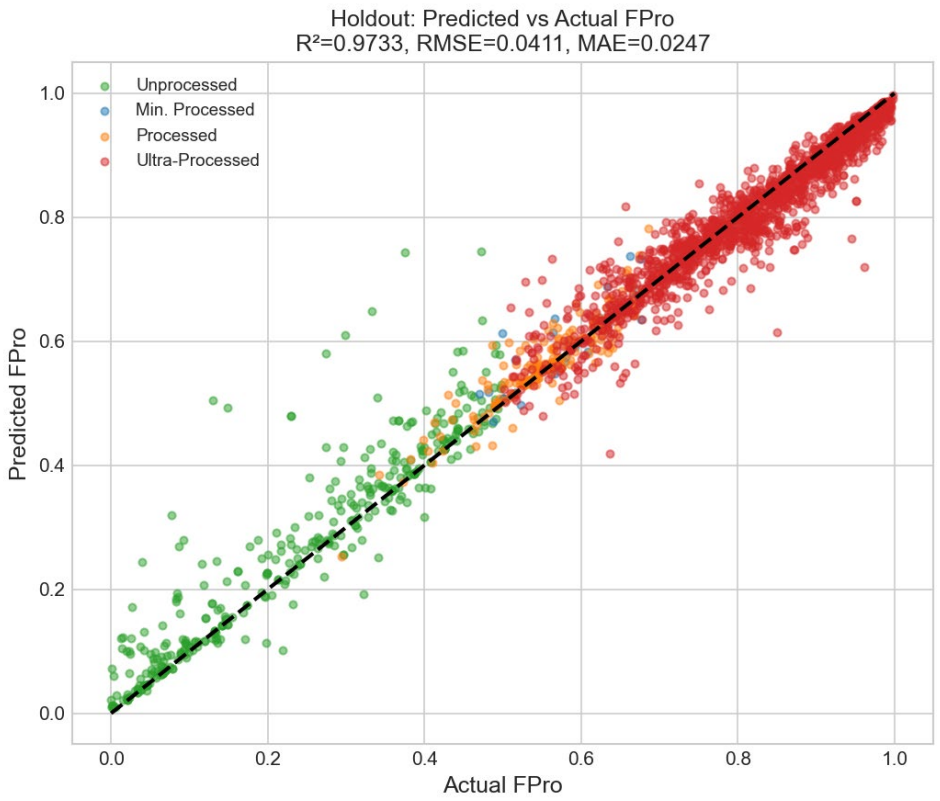
Model	R^2	RMSE	MAE
1. Ridge (baseline)	0.4953	0.1824	0.1408
2. RF (tuned)	0.9756	0.0401	0.0253
3. GB (tuned)	0.9803	0.0359	0.0212

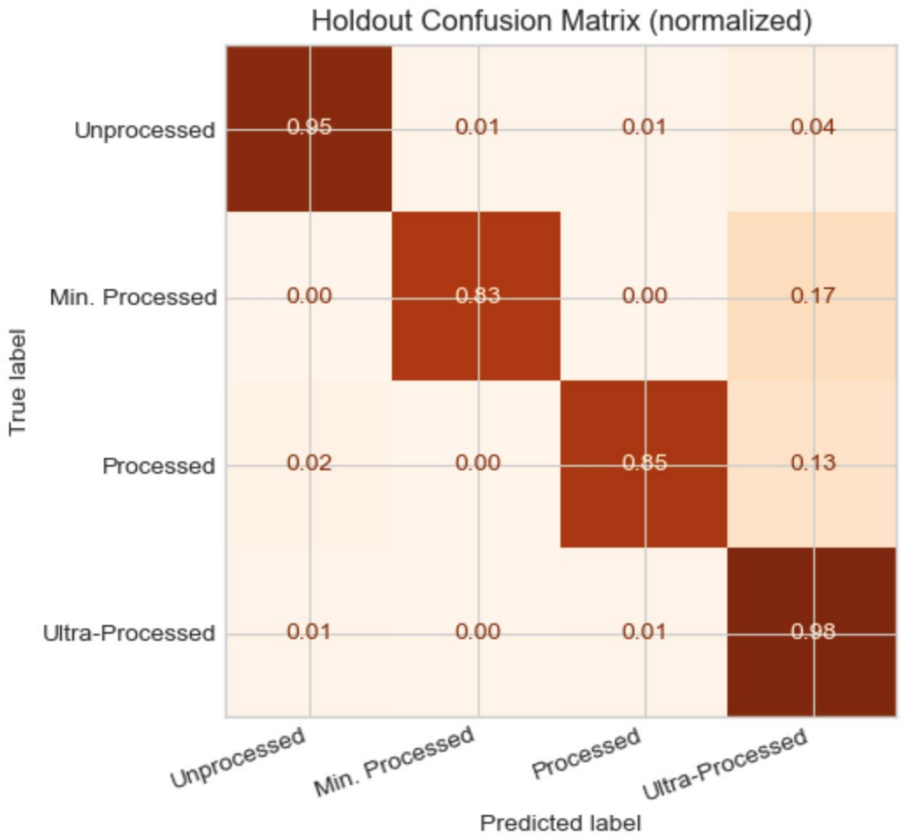
Table 2: Classification Model Comparison

Model	Accuracy	F1 (Macro)	F1 (Weighted)
1. Logistic Reg. (baseline)	0.7465	0.5617	0.7740
2. RF (tuned)	0.9623	0.9048	0.9621
3. GB (tuned)	0.9653	0.9027	0.9647

- In both cases: Random Forest ≈ Gradient Boost > Ridge/Logistic
- Chose RF because of robustness, faster training, minimal hyperparameter tuning, less prone to overfitting, works better with noisy datasets

Model Performance (Grocery Database Holdout)



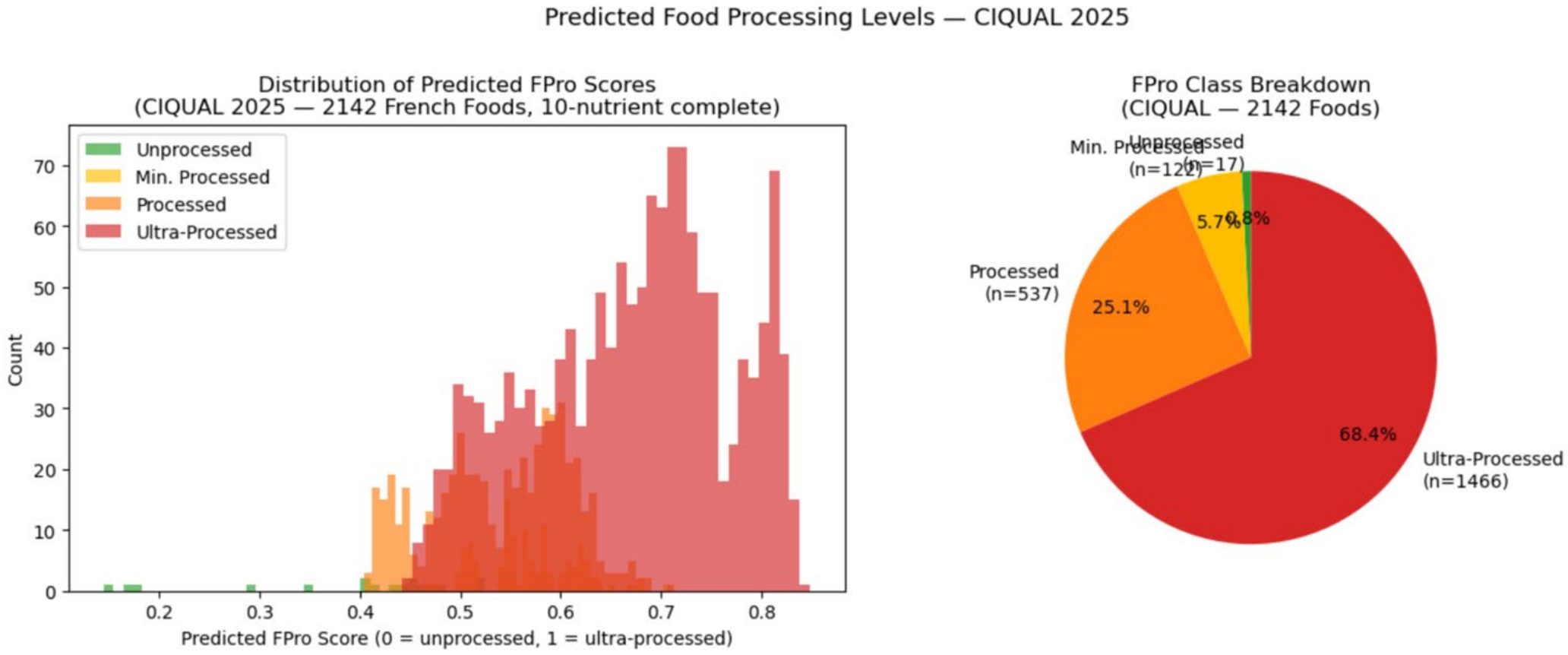


Regression (FPro 0 → 1): R^2 = 0.973, RMSE = 0.041, MAE = 0.025

Classification (4 NOVA classes): Accuracy = 95.9%, Macro F1 = 0.904

10-feature portable model (drop Vitamins A, C): R^2 = 0.974, Macro F1 = 0.891

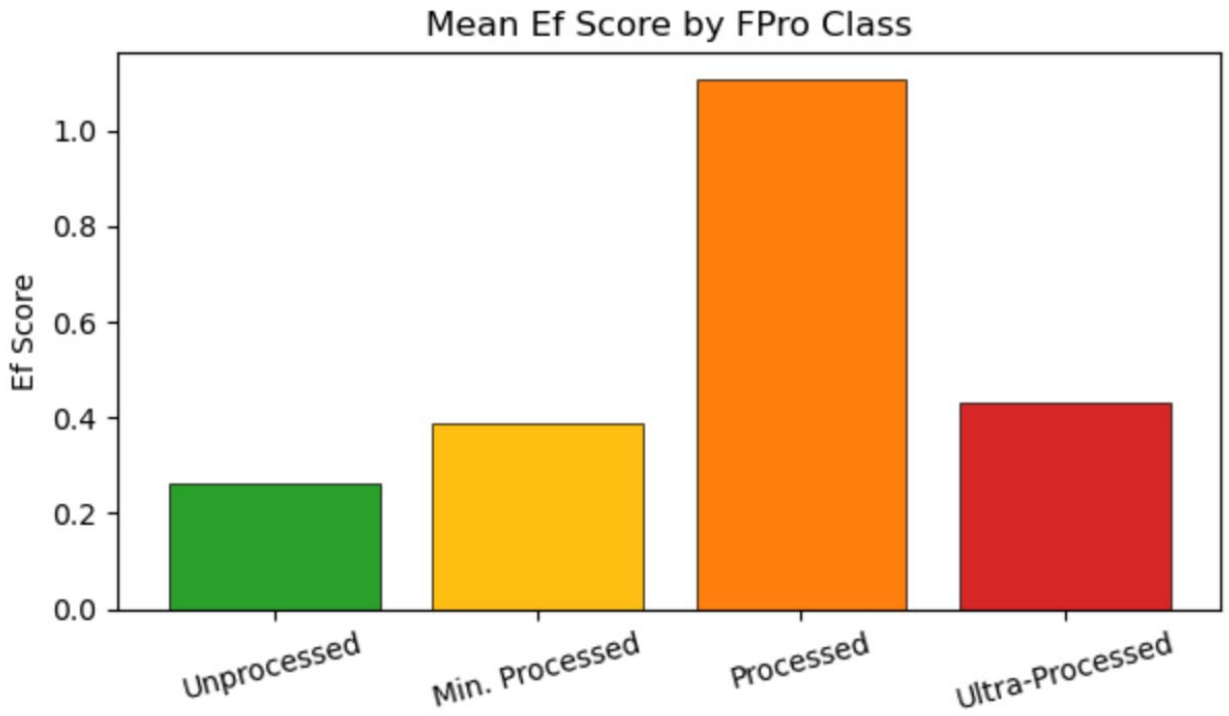
Apply model to CIQUAL (French dataset)



- US model applied to CIQUAL 2025 – no retraining
- 10 features only (Vitamins A, C dropped since sparse in CIQUAL)
- Result: Most CIQUAL foods predicted Ultra-Processed
- Partly real (cereals, beverages, prepared dishes), partly domain shift: US market is 73% UP

Processing impact on environment

- "Unprocessed": **raw meat, fish, dairy** → *very high* GHG emissions from livestock and large land use, but low FPro scores and low Ef score.
- "Ultra-Processed": **grain, sugar, oil-based** (cereals, bread, confectionery) → *lower* climate and land footprint per 100g than animal products.
- **Environmental surprise:** "Processed": **Cheese, cured meat, smoked fish** → HIGHEST footprint (animal raw material + industrial processing)



Conclusions

- Lower processing need not equal lower environmental impact
- Analysis suggests that “Animal vs. plant” is more indicative of environmental impact
- **Moral**: What we eat matters more (at least for the environment) than how processed it is

Future directions

- Linking food processing scores directly to health outcomes
- Effects of different popular diets on environment and health outcomes